Analytics of Unstructured Data, MSBA&AI Fall 2026
Group Assignment #1

Python Notebook due on Canvas by 11:59 p.m. on 9/17 (new due date)

In this assignment you have been hired as an analytics consultant by JD Power and Associates, which wants a competitive analysis of the entry-level luxury car market in the USA. The discussion data have already been scraped from Edmunds.com and will be provided to you in Edmunds.csv; your group does not need to scrape or collect any data. Your job is not simply to produce numbers, but to make defensible analytical choices and turn the social-media conversations into useful competitive insight.

Use of LLMs. You may use LLMs for coding and analysis. However, an LLM-generated answer is not evidence that the analysis is correct. I am deliberately leaving several methodological decisions open. You are expected to decide how to solve them, explain your logic in the submission notebook, validate important steps, and discuss what could go wrong. A technically polished notebook with little evidence of your own reasoning will receive substantially less credit.

Validation. For calculations that can be checked on a small example, create a toy data set and verify the result manually. For semantic tasks, inspect a reasonable sample of classifications/extractions and report the kinds of errors you find. Do not assume that either NLP or an LLM is automatically correct.

Note: You do not have to perform sentiment analysis for this assignment. Unless your method requires otherwise, you may treat mentions as non-negative.

Task A: Discover the important entities and attributes. Without relying on a supplied brand/model dictionary, identify the 10 most important automobile brands in the corpus and the important product attributes being discussed. Your analysis should recognize that consumers may refer to a brand through model names, abbreviations, variants, or other forms. Use an appropriate combination of advanced NLP and LLM methods to discover and reconcile these references. You must decide what should be automated, what requires semantic interpretation, and how to define 'important.' Report the final brands and attributes with their frequencies, and explain and justify the key choices that produced them.

Task B: Establish a simple NLP-based competitive baseline. Using a transparent lexical/NLP approach, calculate lift ratios among the 10 brands identified in Task A. Define what constitutes a brand mention and a co-occurrence, and be consistent about the unit of analysis. Show the lift matrix and explain any normalization or entity-resolution choices that materially affect the results.

Task C: Recalculate lift using LLM-based semantic attribution and reconcile the results. Develop an LLM-based approach that uses the meaning and context of a post, rather than lexical matching alone, to determine the brand references and relationships that should enter the lift calculation. Recalculate the brand-to-brand lift values and compare them with the simpler NLP-based values from Task B. Quantify the agreement/disagreement, investigate the largest differences using representative posts, and explain why the two approaches differ. The objective is not to demonstrate that an LLM is necessarily 'better,' but to determine whether the additional semantic interpretation changes the measurement or managerial conclusions enough to matter.

Task D: Map the competitive structure. Create an MDS map of the major brands using an appropriate transformation of your preferred association measure. Add clustering to the MDS map for better interpretation. Explain the major competitive relationships visible in the map and whether those conclusions are sensitive to the NLP-versus-LLM measurement choice.

Task E: Analyze brand-attribute positioning. Using five important brands and five important attributes from your earlier analysis, determine which attributes are most strongly associated with which brands. Choose and justify an association approach that is consistent with your earlier measurement decisions. Pay particular attention to attribution: when several brands and attributes occur in one post, determine how you will establish which attribute refers to which brand. Explain what the resulting positioning says about the market.

Task F: Identify the most aspirational brand. Determine which brand is most aspirational in the sense that people actually want to buy or own it. You must define and operationalize 'aspirational' from the text rather than simply search for one keyword. Describe your method, validation, result, and the business implications.

Task G: Advise the client. Synthesize the evidence from the competitive, attribute, and aspiration analyses into a small number of actionable recommendations for JD Power and its clients. Distinguish conclusions that are robust from conclusions that depend heavily on methodological choices.

What your notebook must show

- Your code and the main outputs needed to support your conclusions.
- Your methodological reasoning: for each major analytical choice, state what you did, why you did it, and at least one reasonable alternative you considered.
- Evidence of validation, including toy-data checks where appropriate and manual inspection of a sample of semantic classifications/extractions.
- Frequency tables for the major brands and attributes, both lift matrices, the comparison/reconciliation of the two lift approaches, the MDS map, and the details of your aspiration analysis.
- For every LLM-based step, report the model used, number of API calls, input tokens, output tokens, total tokens, and approximate API cost. Explain whether the additional semantic intelligence was worth the cost.
- A concise discussion of limitations and the analyses you would trust least.
- Team member names inside the notebook, not as part of the file name.

The most important part of this assignment is not whether you can persuade an LLM to produce code. It is whether you can make, test, explain, and defend the analytical decisions that turn messy text into credible managerial insight.

Purpose. This guide clarifies how to exercise judgment about LLM use and what evidence your notebook should contain. It does not prescribe a single correct workflow. Selecting, testing, and defending an analytical approach remains a central part of the assignment.

The governing principle. Use an LLM when the task genuinely requires interpretation of meaning or context. Use transparent code for operations that a computer can perform deterministically, including counting, aggregation, formulas, transformations, MDS, and clustering. An LLM may help you write or debug code in either case, but you remain responsible for understanding and validating what the code does.

1 Guidance by assignment task

Task	Suitable role for an LLM	What should remain transparent and student directed
A	Entity discovery and semantic reconciliation of brand, model, and attribute references.	Define importance; generate initial candidates; consolidate entities; count frequencies; validate the final dictionary or classification.
B	Optional help with coding or debugging only.	Use lexical or NLP rules to define mentions and co-occurrence, calculate lift, and verify the formula on toy data.
C	Semantic identification of brand references and relationships within each post.	Specify the output schema and prompt; save classifications; compute lift in Python; compare with Task B; inspect disagreements.
D	Optional assistance interpreting a completed map, subject to your verification.	Choose and justify the association-to-dissimilarity transformation; run MDS and clustering in Python; assess sensitivity.
E	Brand-attribute attribution when proximity or keywords cannot resolve which attribute describes which brand.	Select five brands and attributes; aggregate results; calculate association measures; inspect attribution errors.
F	Classify contextual evidence of desire or purchase and ownership intention.	Define aspiration before classification; design labels; calculate brand scores; validate samples; consider a transparent or hybrid baseline.
G	Optional assistance organizing a draft or checking clarity.	Decide which findings matter; distinguish robust from method-dependent conclusions; make recommendations supported by displayed evidence.

2 A useful decision test

Before using an LLM, ask: Would a clearly specified rule or standard Python function produce the same answer every time? If yes, prefer the transparent method. If the answer depends on meaning, context, or attribution, an LLM may be justified. Then ask whether its likely improvement is important enough to warrant the added cost, opacity, and validation effort.

Hybrid methods are often appropriate. For example, lexical rules can handle obvious brand mentions while an LLM reviews ambiguous cases. If you use a hybrid method, specify the routing rule and validate both the routed and non-routed cases.

3 What the submitted notebook should contain

Submit one executed Python notebook that allows a reader to follow the chain from raw posts to managerial recommendations. The notebook should show reasoning, code, essential outputs, validation, and interpretation. It should not be a code dump or a polished report with the analytical trail removed.

Opening information

- Assignment title and all team members' names inside the notebook.
- A brief executive summary stating the principal findings and recommendations after the analysis has been completed.
- A short statement of the overall division of labor between NLP, LLM methods, and deterministic Python calculations.

Reproducible setup and data audit

- Imports, package installation if needed, model names, important parameter settings, and random seeds.
- Code that loads Edmunds.csv and reports the number of rows, relevant columns, missing values, duplicates, and other consequential data issues.
- Do not include API keys in the notebook. Use Colab Secrets or another secure method.

A through G analysis sections

- For each task, use a clear heading and begin with the question being answered.
- State what you did, why you chose it, and at least one reasonable alternative you considered.
- Show the code and the main output needed to support the conclusion, followed immediately by your interpretation.
- Keep intermediate data available for reproducibility, but avoid printing thousands of rows or full API responses. Show compact samples and summaries.

Validation evidence

- For lift and other checkable calculations, create a small toy example, calculate the expected result manually, and show that the code reproduces it.
- For LLM classifications or extractions, inspect a reasonable sample and report the sample size, selection method, errors found, and any changes made in response.
- For the NLP-versus-LLM comparison, quantify agreement and investigate representative posts behind the largest differences.
- Include sensitivity checks when conclusions depend on thresholds, transformations, clustering choices, or alternative measurement methods.

Required analytical outputs

- Brand and attribute frequency tables; the NLP lift matrix; the LLM-based lift matrix; and a quantitative reconciliation of the two.
- The MDS map with clusters and an explanation of the principal competitive relationships.
- The five-brand by five-attribute positioning analysis, including how brand-attribute attribution was handled.

- The aspiration definition, classification or measurement method, validation results, brand comparison, and business implications.
- A small number of recommendations that clearly trace back to the evidence.

Closing assessment

- Identify the analyses you trust most and least, with reasons.
- Discuss limitations arising from the corpus, entity resolution, attribution, LLM variability, and the chosen definitions.
- Separate robust findings from conclusions that change materially under reasonable alternative choices.

4 Recommended structure for each major task

Question. What specifically are you trying to determine?

Method and decision. What did you do, why, and what reasonable alternative did you reject?

Code. Show readable code that implements the stated method.

Output. Display the table, matrix, figure, sample, or diagnostic needed to evaluate the result.

Validation. Demonstrate that the calculation or semantic classification works well enough for its intended use.

Interpretation. Explain what the result means for the analytical question and the client.

5 Final notebook check

☐ The notebook runs from beginning to end in a fresh Colab session.

☐ All important outputs remain visible when submitted.

☐ Every major methodological choice includes a rationale and an alternative.

☐ Every LLM-based step includes validation and the required usage and cost statistics.

☐ Tables and figures have readable labels, units, and titles.

☐ The recommendations are supported by results shown earlier in the notebook.

☐ Long prompts, exhaustive output samples, or supporting diagnostics appear in a clearly labeled appendix when they would interrupt the main narrative.

☐ No API keys, passwords, or other credentials appear in the submitted file.

What will receive the most credit. A strong submission will not merely produce sophisticated output. It will make consequential decisions visible, test whether the methods work, explain where they fail, and convert the evidence into credible managerial insight.